# Ntokozo Khanyile

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## EDUCATION

DURBAN UNIVERSITY OF TECHNOLOGY — Durban, KZN

Bachelor of ICT in Internet of Things — Graduation May 2026

**Cumulative GPA**: 73%; B+

**Relevant coursework**: Software Engineering; Operating Systems; Algorithms, Artificial Intelligence; Cloud Computing, Identification Technology

## PERSONAL PROJECTS

SMART APPLICATION SCREENING SYSTEM — Feb 2026

*Python, PostgreSQL, FastAPI, JWT, React, Docker, AWS*

- Designed a smart rule-based system to support administrators in quickly responding to University applications.
- I achieved this by implementing a smart rule-based system that suggests decisions for applications and auto rejects applications that do not meet the minimum requirements. The system suggests approval and admin review for suitable candidates with a reason for the decision.
- The system is a human-in-the-loop system allowing the administrator to override the decision.
- Implemented role based access using JavaScript Web Tokens(JWT).
- Developed the backend REST API using the combination of the modern FastAPI python framework with the powerful PostgreSQL database
- Ran backed tests using Pytest.
- The smart rule-based method will later be replaced by machine learning models.
- The frontend is under development using the React.

NATESA HACKATHON — Oct 2025

*Web Development, PostgreSQL, JavaScript, Teamwork*

- Designed and implemented a web based community platform for the Nazareth Tertiary Student Association(NaTeSA) on a 4-day online team Hackathon.
- I contributed in the design, styling, and functionality of the UI of the platform.
- Awarded 3rd place prize.

## UNIVERSITY PROJECTS

MACHINE FALURE PREDICTION MODEL — Nov 2025

*Python, Anaconda, Jupyter, Machine Learning, Web Development*

- Contributed to data cleaning, feature engineering, model training, and model evaluation to produce a model that helps reduce production downtime by predicting machine failure.

- Collaborated in a team to design and document an end-to-end ML workflow.
- Co-authored a technical pare documenting the project's methodology, results, limitations, and reviewed existing industry research.
- Announced top 10 final year project

SMART WATER METER Nov 2024
*ESPN32, Arduino UNO, IoT*

- Designed and Implemented an IoT powered water metering system in a 5-person team using smart IoT sensors and the ESPN32 microcontroller.
- The project idea was to help building managers and tenants to be able to monitor their water usage digitally in real-time and be able to digitally shut and open water in units without calling a technician.

## TECHNICAL SKILLS

- **Software Development**: Python, JavaScript, FastAPI, React, PostgreSQL, MySQL
- **Data & Machine Learning**: Pandas, NumPy, Scikit-Learn, Anaconda, Jupyter Notebook/Labs
- **Web**: REST APIs, JWT, HTTPS, Web Sockets
- **Cloud & Tools**: Git, GitHub, Docker, Visual Studio Code, Azure Fundamentals, ESPN32, Arduino UNO, Bash, SQL
- **Other**: Adaptability, Problem Solving, System Thinking, Documentation, Team Work, Communication

## Activities

STUDENTS4STUDENTS MOVEMENT Jan 2025
*Founding President*

- Founded a peer to peer non-profit that develops and implement digital tools and strategies to help under resourced learners to have access to university applications and study materials.
- The organisation also coordinates trainings and webinars to assist other organisations adapt to using digital resources to maximise their impact.
- Built healthy relationships with 4 highly reputable organizations locally.

## References

Available upon request